MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 7417

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Magnetron, 10 kw (min), 5500 to 5600 Mc, fixed frequency, pulsed type, air-cooled, integral magnet, unipotential cathode.

PHYSICAL CHARACTERISTICS: (See figure 1).

WEIGHT: 8 lbs. approximate
INPUT CONNECTION: Figure 1
OUTPUT CONNECTION: Figure 1

MOUNTING POSITION: Vertical
SUPPORT: Figure 1
MAGNET ISOLATION: Note 5

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ib	P1	pl	Du	tpc
Units:	a	W	kw	--	us
Maximum:	8	50	50	0.001	1.0
Minimum:	-	--	--	--	0.15

Parameter:	Ef	eb	epy	tk	VSWR	Anode T	Input Bushing Presaurization
Units:	V	kv	kv	sec	--	°C	psia
Maximum:	7	8.2	8	180	1.5:1	100	45
Minimum:	--	--	--	--	---	---	13
	(Note 1)					(Note 2)	(Note 3)

CONDITIONS (Note 4)
TEST 1

Parameter:	Ef	Ib	tk	tp	Du	rrv	epy	VSWR	F
Units:	V	A	sec	us	--	Kv/us	Kv	----	Mc
Maximum:	6.3	4	180	0.22	0.0003	80	7.8	1.1:1	5600
Minimum:	--	---	---	0.18	--	--	7	----	5500
	(Note 16)	(Note 16)							

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required (Note 13)

Section 5 - Preparation for delivery

FSC 5960

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	AQL (Percent Defective)	INSP. LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNITS
	Qualification							
-----	Anode-cathode capacity				C	6	12	uuf
1023	Temperature coefficient	Anode temperature 40 to 70°C, (Note 2)			ΔF/ΔT	--	0.2	Mc/°C
1026	Low temperature operation	(Note 12)			T	-40	--	°C
1026	High temperature operation				T	+75	--	°C
-----	Temperature rise	Anode temperature TA = 25°C Approx. (Note 2)			T	+25	--	°C
	Quality conformance inspection part 1 (Note 14)							
1301	Heater Current	Test 1			If	1.40	1.75	A
----	Standing wave ratio	Test 1						
4218	Fix-tuned frequency	Test 1						
4303	Heater-cathode warmup time	Test 1			tk	--	180	sec
4304	Pulse characteristics	Test 1						
4304	Peak current	Test 1						
4306	Pulse voltage	Test 1						
4307	Power output	Test 1			Po	3	--	W
4308	Spectrum measurement	ib = 3A to 8A (Note 7)			BW	--	14	Mc
4315	Load stability	VSWR 1.5:1, ib = 3A to 8A (Notes 9 and 10)			Missing Pulses	--	1	%
4315	Starting stability	Load stability conditions			Missing Pulses	--	1	%

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	AQL (Percent Defective)	INSP. LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNITS
	Quality conformance inspection, part 2							
----	Shock	G = 15 (Note 15)						
1031	Low frequency vibration	No voltage	6.5	II				
1031	High frequency vibration	No voltage	6.5	II				
4309	Non-oscillating characteristics							
	Minimum VSWR		6.5	II		10:1	--	VSWR
	Position of minimum VSWR	(Note 6)	6.5	II	dia	0.55	1.0	cm
4310	Pulling factor	(Note 8)	6.5	II	ΔF	--	13	Mc
	Quality conformance Inspection, Part 3							
4.7	Life test provisions	Test 1, Group D ib = 5.5A (Note 11)			t	500	--	hrs
4.7.3	Life test end points Power output	Test 1			Po	2.4	--	W
4315	Load stability	VSWR = 1.5:1. ib = 3A to 6A (Notes 9 and 10)			Missing Pulses	--	2	%

NOTES:

1. The maximum value specified herein is for non-oscillating condition. Heater surge current shall not exceed 4 amperes.
2. The temperature shall be measured at the point indicated on Figure 1.
3. The gas used in pressurization shall provide insulating properties at least equal to that of clean, dry air at the pressures indicated herein.
4. These conditions shall apply unless otherwise specified herein.
5. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization. A minimum distance of 4 inches shall be maintained between the tube and ferromagnetic materials or energized magnets.

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NOTES (Continued)

6. The plane containing the voltage standing wave minimum shall be within the limits specified herein when measured from reference plane D in the direction of reference plane B. The position of the minimum VSWR and the VSWR shall be determined at the operating frequency obtained during Test 1.

7. Stability will not be measured under this test. The RF bandwidth shall be within the limits specified when a VSWR of 1.5:1 is introduced in the load at a distance of approximately 0.25 meters from the magnetron coupling flange, the phase being adjusted for the widest spectrum.

8. The pulling measurement shall be made in such a manner that thermal effects will not introduce appreciable errors.

9. No more than the percentage of pulses specified herein shall result in r.f. output pulses each having less than 70 percent of the energy contents of a normal pulse. The test shall be conducted for a period of 5 minutes, and satisfactory operation shall be obtained for each minute of this 5 minute period.

10. Observations shall be made for load phase positions corresponding to maximum power and to minimum power.

11. The tube shall pass all applicable Quality Conformance Inspection Part 1 tests at the end of the Life test specified herein except as modified under Life test and points.

12. The requirements of the starting stability test, with the exception of holding period, shall be the only criterion for this test.

13. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D.C. 20360 and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

14. Quality conformance inspection part 1, shall be a combined AQL (percent defective) of 1 percent and the inspection level shall be II.

15. The magnetron shall be mounted on a test plate and dropped five times on each of three mutually perpendicular axes parallel to the reference planes shown on Figure 1. The shock pulse shall have a duration of approximately 11 ms as measured at the quarter amplitude points of the acceleration shock wave. This is not a destructive test.

16. The test shall be run with Ef set at 6.3 volts. The epy shall be adjusted to provide an ib of 4 amperes.

Review activity:
Navy - SH

User activity:
Navy - WP, MC, CG

Preparing activity:
Navy - SH
(Project 5960-N615(NAVY)

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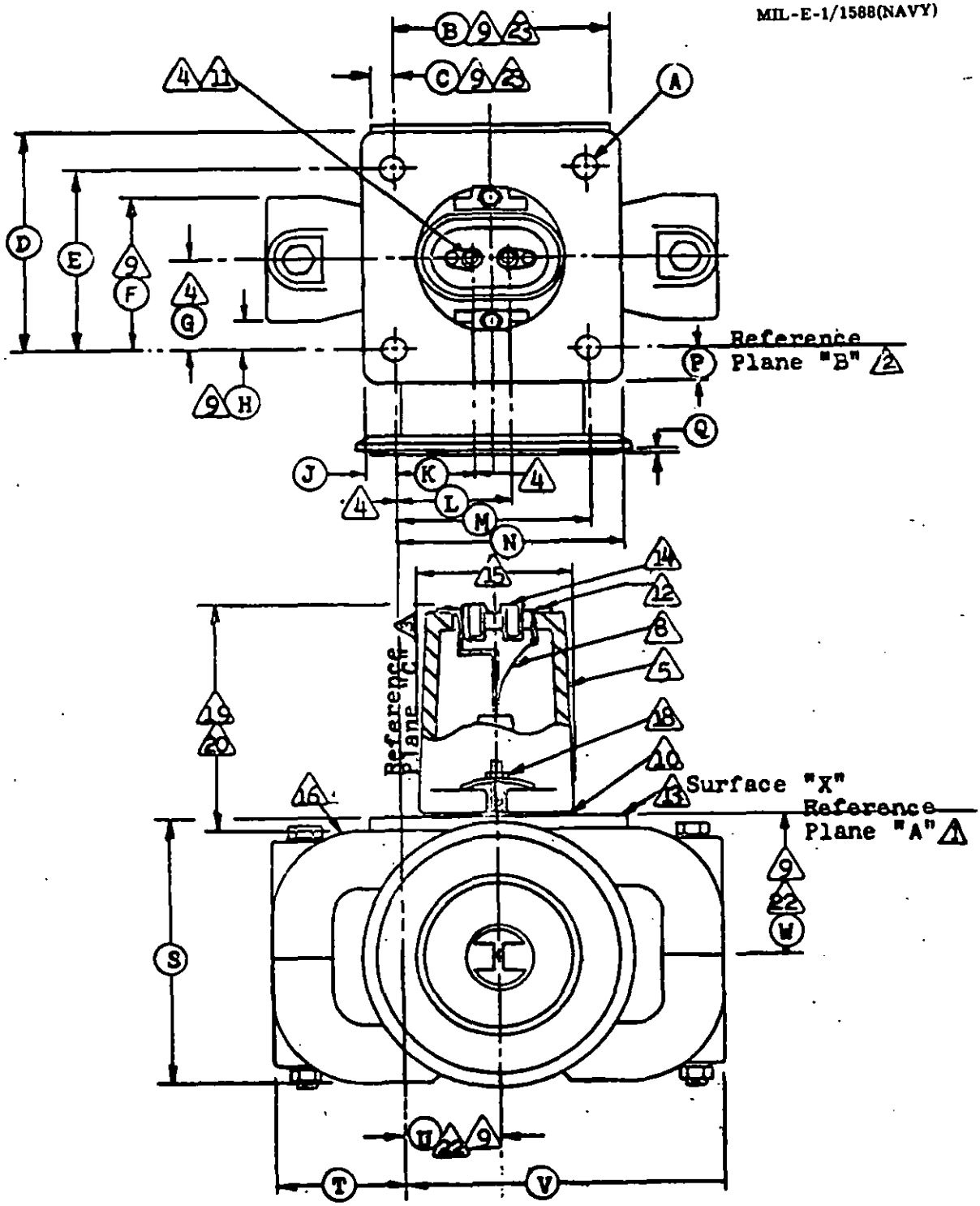


Figure 1 - Outline and physical dimensions of tube type 7417.

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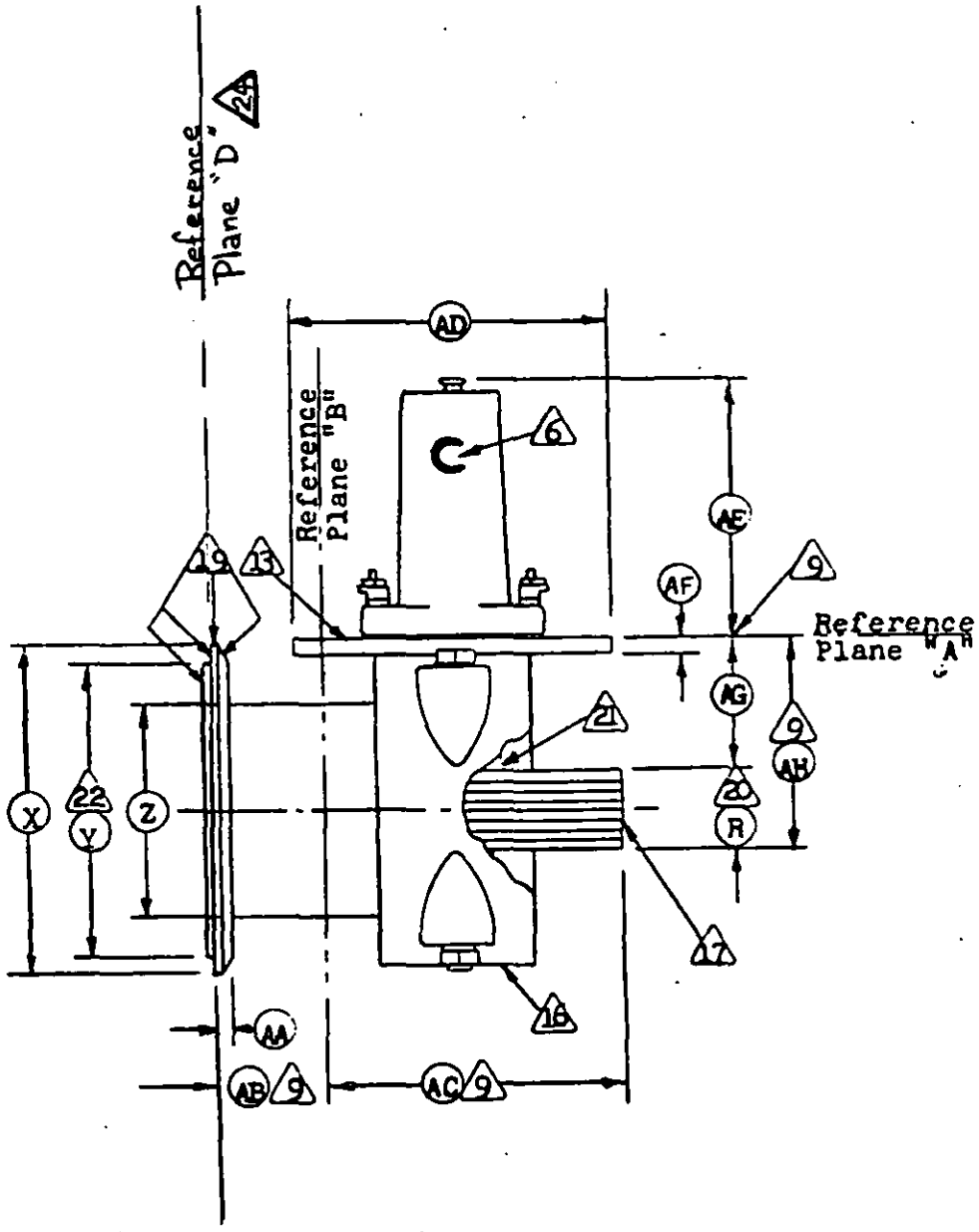


Figure 1 - Outline and physical dimensions of tube type 7417 (Con'td).

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Dimension	AQL (Percent Defective)	INSPECTION LEVEL OR CODE	Minimum		Maximum	
			Inches	Millimeters	Inches	Millimeters
Qualification						
B					2.937	74.40
C					0.312	8.19
D					3.025	86.83
E			2.610	66.29	2.640	67.06
F					2.437	61.90
G			1.302	33.07	1.322	33.58
H			0.187	4.75		
J					0.400	10.16
P					0.400	10.16
R			0.839	21.31	0.939	23.85
S					4.000	101.60
T					2.000	50.80
V					4.625	117.48
X			3.615	91.82	3.635	93.33
Y			3.308	84.02	3.316	84.23
Z			2.365	60.07	2.385	60.60
AA			0.232	5.90		
AC			3.177	80.50	3.197	81.20
AD			3.345	85.00	3.407	86.40
AE			2.940	74.60	2.960	75.20
AF			0.187	4.75		
AG			1.312	33.40		31.30
Quality conformance inspection, part 1						
Q	6.5	II	0.027	0.68	0.037	0.94
U			1.280	32.50	1.344	34.17
W			1.843	46.80	1.907	48.40
AB			1.218	30.90	1.232	31.30
AH					2.562	65.00
Quality conformance inspection, part 2						
A	6.5	II	0.340	8.64	0.350	8.88
K			1.052	26.80	1.072	27.20
L			1.552	39.42	1.572	40.00
M			2.610	66.25	2.640	67.00
N					3.025	76.80

Figure 1 - Outline and physical dimensions of tube type 7417 (Cont'd).

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NOTES:

1. Reference Plane "A" is defined as a plane passing along the face of the mounting plate.
2. Reference Plane "B" is defined as a plane perpendicular to Plane "A" passing through the center of the holes as shown.
3. Reference Plane "C" is defined as a plane mutually perpendicular to Planes "A" and "B" passing through the center of the hole as shown.
4. The jack holes shall be within a radius of .023 of the true location specified but shall be spaced .500 ±.010 with respect to each other. The centerlines of the holes shall be perpendicular to Reference Plane "A" within 3°0'.
5. Nylon, porcelain or approved equivalent.
6. Indicate common cathode connection by letter "C".
7. Surface of plate extending beyond a 1.125 radius from the center of plate, shall be flat within .015.
8. Filament leads shall be flexible and slack.
9. Tolerance includes regular as well as lateral deviations.
10. .031 asbestos gasket (optional).
11. Hex head banana pin .169 ±.005 dia hole.
12. Leads soldered to terminal clips.
13. All solder joints on mounting plate and output section shall be such as to provide an hermetic seal at surface X.
14. Heater connection.
15. Any portion of the assembly extending above this surface shall be within a 1.109 radius of true center of plate.
16. Magnet with protective coating.
17. Radiator fins.
18. #6-32 elastic stop nut .312 across flats.
19. All areas except top of insulator to bottom edge of mounting plate, face, edge, and angle of flange shall be painted with black lacquer.
20. These dimensions apply to radiator size only.
21. Point at which temperature of tube is measured.
22. Refers to centerline of output flange as determined by 3.306 - 3.318 dia.
23. Dimension applies to radiator.
24. Reference Plane "D" is defined as a plane passing through the cutback section of the waveguide.

Figure 1 - Outline and physical dimensions of tube type 7417.

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